

REPORT

Course Code: CSE 4408

Course Name: System Analysis and Design

Lab Number: 6

Project Title: BashayJabo

Group: Les Misérables

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1. Sampling Strategy Design

Populations Identified

We identified three relevant populations pertaining to ride-sharing among IUT students whose behaviors and insights will influence the system's design:

1. Frequently Commuting Students

- Students who travel to and from IUT multiple days a week with or without ride-sharing.
- Their input will help us understand daily needs, route patterns, and transportation preferences.

2. Occasional Travelers

- Students who travel only during longer breaks or on emergency bases, and most likely travel long distances from Dhaka.
- Will help us explore flexibility and one-time needs.

3. Regular Organisers (Informal Ride Coordinators)

- These are students who frequently initiate and coordinate group rides.
- They hold valuable insights into how rides are currently arranged, including scheduling, fare negotiation, and handling last-minute changes.
- Identify pain points in coordinating shared rides informally via messaging.

Chosen Population and Sampling Method

- **Chosen Group:** Regular Organisers
- **Sampling Method:** Purposive Sampling

Justification:

This project seeks qualitative, experience-rich feedback regarding preferences and challenges in group coordination to inform requirement gathering. Since student ride organizers have the most first-hand experience in managing shared rides and they know the process, its common pain points, and their informal workarounds, purposive sampling ensures we gather deep, targeted insights rather than general opinions—the system essentially aims to automate organiser behaviour. Purposive sampling also respects time and resource constraints for our small team, as random sampling would require contacting a much larger base of students with possibly irrelevant experience.

Sampling Procedure

A systematic, step-by-step process will be followed to draw our purposive sample:

1. Define Inclusion Criteria:

- Target IUT students who initiate or organise shared rides at least once or twice per week.

2. Identify Participants:

- Look into IUTian group chats on Facebook, WhatsApp, etc. to identify potential participants.
- Ask peers to recommend known ride coordinators.

3. Choose participants:

- Choose 15–20 participants based on convenience—whoever responds earliest.
- Ensure participants with diverse travel patterns & distance, to capture varied perspectives.

4. Invite Participation:

- Send polite, concise messages explaining the purpose of the system.
- Request their availability for interview, observation, or sharing past ride coordination examples (screenshots, logs, etc.).

2. Investigation Plan for Existing Data

Three Existing Data Sources

1. Social Media Posts (Facebook)

- Type: Qualitative
- Posts on public Facebook groups asking for rides, from which the analysis of real-time responses will reflect the efficiency of such methods.

2. Messaging / Chats

- Type: Qualitative
- Screenshots of ride seeking in personal or group chats, showing transportation, timing, and route details, as well as choice in preferred partners.

3. Phone Calls / IRL Conversations

- Type: Qualitative
- Casual verbal exchanges about who's going when, with whom, and how to split the fare, which highlight the trust-based, informal nature of current coordination.

Selected Data Source: Messaging / Chats

Anonymously shared screenshots of student ride-sharing conversations provided valuable insight into patterns of behavior around commuting coordination. While the exchanges are disorganized at times, they offer an authentic glimpse into the most common method of planning rides among IUT's tech-savvy student population.

Expected Insights:

- Reveal recurring patterns in how and when ride-sharing is initiated.
- Show delays or confusion in ride confirmations (e.g., multiple students replying late or asking repeated questions).
- Highlight the lack of structured templates (e.g., users often forget to mention pickup points or fare), as well as fallback support when schedules change or are delayed (e.g., a user depending on a ride has no backup if their partner decides to skip class).
- From these gaps in manual coordination, the scope for automation is clarified and the implied features requested by students can be derived.

How It Can Inform Requirements/UI Design:

- Demonstrates a need for pre-filled ride request templates within the app.
- Shows importance of real-time availability indicators or auto-matching with peers based on routine schedules.
- To complement the trust-based search for ride partners, a feedback or rating system for users will ensure mutual respect and prevent misbehaviour.
- Suggests features like instant ride confirmation, in-app fare splitting, or status update push notifications when plans change.

Limitations:

- Chat logs are unstructured and informal, with inconsistent language and missing details or context (e.g., inside jokes or colloquialisms that need to be deciphered).
- Hard to quantify behavior without additional manual interpretation, automated analysis is insufficient.
- Cannot analyse user fallback workarounds since collected data may not be arranged or interpreted in that manner.

3. Observation Planning – Analyst’s Playscript

User Role Selected: Student Commuter (Ride Initiator)

This role is essential because the ride initiator’s actions directly influence how quickly, clearly, and efficiently a shared ride is arranged. Observing this role helps uncover friction points and automation opportunities.

Playscript: Sequential Tasks Performed by the Student (Ride Initiator)

1. Opens group chat

- The student opens Messenger or WhatsApp on their phone or laptop to begin organising the ride. They mostly use departmental group chats or groups with peers who have similar schedules.

2. Texts desired ride time/location/vehicle

- Types out a message indicating destination, preferred time, number of seats, and sometimes fare estimate. This is usually informal with no standard format.
- Most common user preferences can be analysed from this.

3. Waits for responses from others

- Waits for others to respond with interest. This stage often causes delays, as replies may come at irregular intervals or require clarification.
- Response rate and user reaction to delays can be tracked through this.

4. Confirms participant and fare

- Reviews responses, selects who to include in the ride, and sometimes resolves scheduling or payment conflicts manually.
- Informal confirmation and fare splitting practices can be observed.

5. Later updates & follow up

- Posts a follow-up message confirming time, pickup point, and fare split.
- If someone cancels or is late, the initiator either reposts for new participants or changes the timing, often repeating earlier steps.

Insights and Opportunities

- This process is entirely informal and time-consuming, with high chances for miscommunication or last-minute disruptions.
- Automating key actions (e.g., ride creation templates, real-time confirmations, pre-set user preferences) could greatly improve speed and reduce confusion.
- Adding structured fields (e.g., time, location, fare, participants) in a mobile app would reduce manual effort and errors.

4. Observation Planning – STROBE Analysis

Key Organizational Themes from Lab 5

- **Informal Coordination Culture**
 - Ride-sharing is managed through informal platforms like Messenger or WhatsApp.
 - There is no centralized coordination or scheduling tool, making the process time-consuming and inconsistent.
- **Cost-Sensitivity**
 - A primary motivation for ride-sharing is splitting fares.
 - Fare handling is manual and often leads to confusion or discomfort.
- **Last-Minute Uncertainty**
 - Ride plans frequently change due to class timing, traffic, or personal emergencies.
 - This results in missed rides or long waiting times, especially during peak hours.
- **Trust and Safety Concerns**
 - Students prefer to share rides with known or filtered peers (same department, gender, mutual friends).
 - These filters are difficult to apply or enforce in the current system.

Workspace STROBE Assessment

Since there is no dedicated office/workspace for students who want to plan shared rides, our STROBE analysis is adapted to the informal gathering points around campus where coordination takes place.

1. Office Location

- Students typically plan rides in informal settings like dorm rooms, the CDS cafeteria, or while walking to/from class. There is no dedicated “workspace” for ride coordination, which contributes to distraction and inconsistent communication. From this it can be interpreted that there is a lack of defined “coordination hub” which leads to chaos and inefficiency.

2. Desk Placement

- No fixed desk is used. Coordination happens on-the-go, often groups form loosely around pickup points like hall gates, main gate, around the IUT bus, etc. There is no structured flow or local point. This suggests a preference for mobile-first interfaces that support multi-tasking and quick access.

3. Stationery Equipment

- All coordination relies on personal phones and verbal exchanges. This implies that coordination is manual, unstructured, through informal messaging/ calls/ conversations. Any proposed system should minimize typing and use structured input fields or voice/text templates.

4. Props

- Primary props include smartphones, earphones, bags. There is no centralized app or interface to coordinate effectively. Coordination tools should be lightweight, mobile-compatible, and integrate with calendars or reminders.

5. External Information Sources

- External sources are minimal, no visible use is seen. Most students rely on each other for information. System design should integrate external APIs for ride estimates or map-based features for smoother coordination.

6. Office Lighting and Color

- Sweltering heat and outdoor crowding can be seen. The environment is uncomfortable and stressful, not ideal for planning.

7. Clothing/Authority

- No uniform or role-based attire; coordination is casual, peer-led. This can become confusing if there is little organisation. The system should reflect equal access and decentralized decision-making, while allowing users to filter by preferred groups (e.g., same department/gender).

5. Synthesis with Interactive Methods

Anecdotal List Derived from STROBE

1. **“We usually gather at CDS”**
 - Common physical space for ride coordination.
2. **“Two lines leading out from CDS... to IUT bus, to main gate”**
 - Movement patterns suggest predictable pickup locations.
3. **“Asking each other about plans”**
 - Students rely on peer-to-peer communication via chat or verbal exchange.
4. **“We have to find people ourselves”**
 - No automation; students manually search for co-riders.
5. **“We use Uber on most days”**
 - External services are used when internal coordination fails or isn’t available.
6. **“We often stand in the heat”**
 - Environmental discomfort while waiting or coordinating rides.
7. **“No leadership mentioned”**
 - Ride coordination is decentralized and peer-led.

Comparison with Interactive Methods (Lab 5 Interviews/Surveys)

1. **“We usually gather at CDS”**
 -  **Confirms:** CDS was consistently mentioned in interviews as a coordination hub.
2. **“Two lines leading out from CDS... to IUT bus, to main gate”**
 -  **Modifies:** Survey responses discussed routes but not spatial patterns—this adds new clarity.
3. **“Asking each other about plans”**
 -  **Confirms:** Chat-based informal planning was widely reported in surveys.
4. **“We have to find people ourselves”**
 -  **Supplement:** Adds to interview feedback on the need for automation in finding partners.
5. **“We use Uber on most days”**
 -  **Cue:** Implies some students are familiar with app-based transport, suggesting external service integration is viable.

6. “We often stand in the heat”

-  **Confirms:** Matches complaints in interviews about waiting outside with no confirmation system.

7. “No leadership mentioned”

-  **Supplement:** Supports the trust-based, peer-coordinated ride culture identified earlier.

Conclusion:

Through unobtrusive methods such as STROBE and playscript observation, we gained valuable insights into the informal, trust-based nature of ride-sharing coordination among IUT students. Comparing these with interactive methods from Lab 5 confirmed key pain points, manual communication, lack of structure, and timing uncertainty. These findings highlight the need for a mobile-first, user-friendly system that supports structured ride requests, peer filters, and integration with external services.